In [1]:
```
s = {}
s
```

Out[1]:
```
{}
```

In [2]:
```
type(s)
```

Out[2]:
```
dict
```

In [3]:
```
s = set()
s
```

Out[3]:
```
set()
```

In [4]:
```
type(s)
```

Out[4]:
```
set
```

In [12]:
```
s1 = {150, 85, 80, 2, 89, 60}
s1
```

Out[12]:
```
{2, 60, 80, 85, 89, 150}
```

In [13]:
```
type(s1)
```

Out[13]:
```
set
```

In [14]:
```
len(s1)
```

Out[14]:
```
6
```

In [15]:
```
s1.add(34,67)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[15], line 1
----> 1 s1.add(34,67)

TypeError: set.add() takes exactly one argument (2 given)
```

In [16]:
```
s1.add(65)
```

In [17]:
```
s1
```

Out[17]:
```
{2, 60, 65, 80, 85, 89, 150}
```

In [18]:
```
s2 = s1.copy()
```

In [19]:
```
s2
```

Out[19]:
```
{2, 60, 65, 80, 85, 89, 150}
```

In [20]:
```
s2[0]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[20], line 1
----> 1 s2[0]

TypeError: 'set' object is not subscriptable
```

In [21]:
```
s2
```

Out[21]:
```
{2, 60, 65, 80, 85, 89, 150}
```

In [22]:
```
s2[:]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[22], line 1
----> 1 s2[:]

TypeError: 'set' object is not subscriptable
```

In [23]:
```
s3 = {'b', 4+2j, 8.3, False}
s3
```

Out[23]:
```
{(4+2j), 8.3, False, 'b'}
```

In [24]:
```
print(s1)
print(s2)
print(s3)
```

```
{80, 65, 2, 85, 150, 89, 60}
{80, 65, 2, 85, 150, 89, 60}
{8.3, 'b', (4+2j), False}
```

In [25]:
```
s1
```

Out[25]:
```
{2, 60, 65, 80, 85, 89, 150}
```

In [26]:
```
s1.pop()
```

Out[26]:
```
80
```

In [27]:
```
s1.pop(2)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[27], line 1
----> 1 s1.pop(2)

TypeError: set.pop() takes no arguments (1 given)
```

In [28]:
```
s1
```

Out[28]:
```
{2, 60, 65, 85, 89, 150}
```

In [30]:
```
s1.discard(140)
```

In [31]:
```
s1.remove(140)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[31], line 1
----> 1 s1.remove(140)

KeyError: 140
```

In [32]:
```
s1
```

Out[32]:
```
{2, 60, 65, 85, 89}
```

In [33]:
```
s1.discard(89)
```

In [34]:
```
s1
```

Out[34]:
```
{2, 60, 65, 85}
```

Set Operatiom

In [35]:
```
f1 = {1,2,3,4,5}
g1 = {4,5,6,7,8}
h1 = {8,9,10}
```

In [36]:
```
f1.union(g1)
```

Out[36]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [37]:
```
f1.union(g1, h1)
```

Out[37]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

In [38]:
```
f1 | g1
```

Out[38]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [39]:
```
g1|h1
```

Out[39]:
```
{4, 5, 6, 7, 8, 9, 10}
```

In [40]:
```
f1|h1
```

Out[40]:
```
{1, 2, 3, 4, 5, 8, 9, 10}
```

In [41]:
```
f1|g1|h1
```

Out[41]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

intersection

In [42]:
```
print(f1)
print(g1)
print(h1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [43]:
```
f1.intersection(g1)
```

Out[43]:
```
{4, 5}
```

In [44]:
```
f1.intersection(h1)
```

Out[44]:
```
set()
```

In [45]:
```
f1&g1
```

Out[45]:
```
{4, 5}
```

In [46]:
```
g1&h1
```

Out[46]:
```
{8}
```

In [47]:
```
f1&h1
```

Out[47]:
```
set()
```

In [48]:
```
f1 = {1,2,3,4,5}
g1 = {4,5,6,7,8}
h1 = {8,9,10}
```

In [49]:
```
f1.difference(g1)
```

Out[49]:
```
{1, 2, 3}
```

In [50]:
```
g1-h1
```

Out[50]:
```
{4, 5, 6, 7}
```

In [51]:
```
print(f1)
print(g1)
print(h1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [52]:
```
f1.difference(g1)
```

Out[52]:
```
{1, 2, 3}
```

In [55]:
```
f1.symmetric_difference(g1)
```

Out[55]:
```
{1, 2, 3, 6, 7, 8}
```

In [53]:
```
f1^g1
```

Out[53]:
```
{1, 2, 3, 6, 7, 8}
```

In [54]:
```
f1^g1^h1
```

Out[54]: {1, 2, 3, 6, 7, 9, 10}

In [56]:
```
print(f1)
print(g1)
print(h1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [57]:
```
f1.difference(g1)
```

Out[57]: {1, 2, 3}

In [58]:
```
f1
```

Out[58]: {1, 2, 3, 4, 5}

In [59]:
```
f1.difference_update(g1)
```

In [60]:
```
f1
```

Out[60]: {1, 2, 3}

In [62]:
```
print(f1)
print(g1)
print(h1)
```

```
{1, 2, 3}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

Superset|subset|disjoint

In [63]:
```
s5 = {11,12,13,14,15,16,17,18,19}
s6 = {13,14,15,16,17,18}
s7 = {10,20,30,40}
```

In [64]:
```
s5.issuperset(s6)
```

Out[64]: True

In [65]:
```
s6.issuperset(s5)
```

Out[65]: False

In [66]:
```
s5 = {11,12,13,14,15,16,17,18,19}
s6 = {13,14,15,16,17,18}
s7 = {10,20,30,40}
```

In [67]:
```
s6.issubset(s5)
```

Out[67]: True

In [68]:
```
s7.isdisjoint(s5)
```

Out[68]: True

In [69]:
```
s7.isdisjoint(s6)
```

Out[69]: True

In [70]:
```
s5 = {11,12,13,14,15,16,17,18,19}
s6 = {13,14,15,16,17,18}
s7 = {10,20,30,40}
```

In [71]:
```
s5.isdisjoint(s6)
```

Out[71]: False

In [78]:
```
a3 = {11,12,13,14,15,16,17,18,19}
b3 = {13,14,15,16,17,18}
c3 = {10,20,30,40}
```

In [79]:
```
b3.issubset(a3)
```

Out[79]: True

In [80]:
```
b3.isdisjoint(c3)
```

Out[80]: True

In [ ]: